When Do Parameter Budgets Misprice CPU Inference? A Pre-registered Limit Study of Byte-Traffic Pruning Allocation

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Abstract

Pruning budgets are almost always stated in parameter count, yet CPU decode cost is plausibly driven by the bytes moved per token: weight bytes streamed each step plus KV-cache bytes read at the occupied context. We test, under a pre-registered protocol, whether allocating a $\sim$25% structured prune of Llama-3.2-3B by an analytical per-token byte-traffic model beats parameter-count allocation, at fixed CPU thread caps (2 and 4 threads, both pinned to 2 cores). Eleven allocations search one shared unit space (layers, FFN channels, KV groups) and differ only in the cost model choosing the cut. The tightened evidence does not support the negative result the protocol was built to deliver. A teacher-forced validity check of every exported variant (Amendment A3) passes 1 of 11: only V-depth-lite's GGUF stays within the frozen export-fidelity tolerances, so the pre-registered 10-variant allocation contest (six stage-1-valid plus four amendment variants) cannot be fitted on a D4-valid primary set, and the registered head-to-head has both arms invalid. The frozen local contest over the original six stage-1-valid variants, reported with that caveat: the traffic model does not beat the parameter model by the registered margin at either cap (ΔAIC $+2.21$ at 2 threads, -19.11 at 4), and the predeclared byte allocation does not reliably meet the registered standard (paired ratios 0.917 / 0.986 at 2 threads, 0.818 / 0.817 at 4), though in a post-hoc sensitivity excluding the saturated 16384 context – an exclusion requested in review, not pre-registered – it does win the 4-thread contest. Quality is a context axis only: the frozen quality scorer is void (it scores raw-completion continuations, not answers), and the Amendment-4 first-answer re-score of the frozen outputs gives BASE 85.00 (34/40) and the sole D4-valid variant 55.00 (22/40). The contribution is therefore a boundary, drawn twice over: 10 of 11 allocations fail the export-fidelity standard under the joint prune-and-quantise pipeline (the two steps are applied jointly and never separated), and the traffic model's own predicted speedups run systematically ahead of what was measured. Traffic-aware pruning as such sits outside it — pruning by *measured* latency is untouched by anything we report. Scope: every timing comes from a shared 2-core virtual machine (tenant-load variance is a formal limitation); the outcome is CPU time, not energy.

1 Introduction

Structured pruning removes whole layers, channels, or attention components so that a smaller model runs faster on real hardware. The budget for such a prune is conventionally a parameter count: remove $x\%$ of the weights. Parameter count is, however, a proxy. At batch size 1, autoregressive decode is widely treated as memory-bound: each step must stream the model's weights once and read the accumulated key–value (KV) cache, whose size grows with context. A cost model based on per-token byte traffic — weight bytes streamed per step plus KV bytes read at the occupied context — is therefore an attractive alternative: it prices KV-heavy allocations differently from parameter-heavy ones, and it predicts that two allocations with identical parameter counts can have different decode costs, especially at long contexts.

This intuition has been used to *analyse* inference [Bi et al., 2026] and named as an untested extension by pruning work that itself budgets in parameters [Wang et al., 2026], but we find no prior controlled test of the allocation question (literature search, October 2026): *within one matched-budget pruning family, does choosing the cut by byte traffic produce a faster model, and does the traffic model predict held-out allocations' speed better than parameter count?*

We answer with a pre-registered, single-model, single-runtime experiment in which the two cost models are the only things that differ between arms. Export-validity failures restrict the results to a subset of the design space; this paper reports that narrower result honestly:

1. **Validity.** A teacher-forced check of every variant's exported model passes 1 of 11 (Section 4.1). The pre-registered allocation contest cannot be fitted on a D4-valid primary set; the registered head-to-head has both arms invalid.

2. **Frozen contest, with caveat.** On the original six stage-1-valid variants, the traffic model does not out-predict the parameter model (Section 4.3), and the predeclared byte allocation fails the pre-registered $\geq$10% paired-advantage criterion at both thread caps (Section 4.2). Five of those six fail the amended (D4) validity check on rebuilt models, so the contest is frozen evidence with that caveat.

3. **Model calibration.** Where the traffic model predicts 23–33% advantages at cap 2, observation delivers 1–8%; the decision rule shares the blame, but the model itself is faulty.

4. **Quality as context only.** The quality evaluation ran; its frozen scorer is void, and the Amendment-4 first-answer re-score is reported as a context axis, never a verdict criterion (Section 4.6).

We also report selection regret descriptively (Section 4.5): the byte-greedy pick was in raw terms a decent pick, which makes the model's inaccuracy more informative.

2 Related work

Measured-latency pruning. ZipLM prunes by target-runtime latency lookup tables rather than by size [Kurtic et al., 2023]; HAPE ranks structures by measured on-device latency differences for Llama-2-7B on a Xeon CPU, the closest deployment precedent to ours [Zhao et al., 2025]. Both sidestep analytical cost models altogether, and neither asks whether an analytical traffic model adds predictive value over parameters on unseen allocations. **Structured pruning methods.** LLM-Pruner removes coupled structures by gradient-estimated importance and retrains [Ma et al., 2023]; SliceGPT cuts dimensions in a rotated space [Ashkboos et al., 2024]; Wanda prunes by weight magnitude times activation norm [Sun et al., 2024]. Our variants span the same axes (layers, FFN width, KV heads) at a matched $\sim$25% removal, but this paper proposes no pruning method: the treatment is the cost model that chooses the cut. **Joint FFN–KV allocation.** GRASPrune allocates a single budget jointly over FFN channels and KV groups, but states that "we use parameter count as the budget proxy"; a bandwidth-aware cost is proposed in its §3.1 and named as a limitation, not evaluated [Wang et al., 2026]. Our experiment is, to our knowledge, the controlled test of that extension. **KV-targeted pruning.** KVPruner prunes KV channels and reports that equal parameter counts can yield different speeds, but its inference-overhead evaluation disables caching, so its numbers do not speak to cached decode [Lv et al., 2025]. **Traffic and roofline models.** RooflineBench defines per-token traffic as weight bytes plus active KV reads/writes — the formula family we use — and benchmarks it across on-device hardware including Llama-3.2-1B, without a pruning-allocation test [Bi et al., 2026]. Analytical models separate weight and KV memory costs [Pope et al., 2023]; CPU characterisation and kernel-level simulation both find that a single bandwidth constant cannot capture decode performance [Na et al., 2024, Yao et al., 2026]. **This venue.** EMC^2's own program prices the same quantities: KV-cache retention as a carbon tradeoff [Bassi et al., 2026]; NF4 dequantisation as a first-order cost [Qi et al., 2026]; efficiency by composite index [Parihar, 2026]; operation fusion, which changes performed traffic, as a speedup [Salmani & Soloveychik, 2025]. Our design asks whether the simplest analytical traffic model earns its keep for one decision: where to cut. The topic list names "limit studies" and "bottleneck analysis" explicitly; this paper is one.

3 Method

3.1 Variants and cost models

We prune Llama-3.2-3B-Instruct (GGUF Q4_K_M; KV cache fp16) by $\sim$25% of parameters. Eleven allocations search one shared search space (whole layers, FFN channels, KV groups); allocations differ only in the cost model that prices each unit: parameter count, marginal byte traffic ΔB at the design contexts, or a mix of the two. Final parameter counts span 2.398–2.442B (1.84% spread); tensor-payload bytes for the original seven span 1.526–1.552 GB; parameter count enters all analyses as a covariate, so exact equality is not load-bearing. SIMD-alignment constraints bind all variants (FFN multiples of 256, hidden widths multiples of 64, head dimension 128). The six originally stage-1-valid variants are the frozen contest set; the four amendment variants (A1) extend it to ten for the amended contest.

Both candidates are produced by applying their cost model to the *identical* candidate space, so their differing composition is the treatment, not a confound: the parameter model spends the whole budget on FFN width (layers and KV heads are expensive in parameters per unit), while the byte model spends it on layers and KV heads (expensive in traffic per unit). The remaining variants span intermediate selections of the same space, so the prediction contest (Section 4.3) is over allocations, not only over the two endpoints.

Variant	Layers	KV-heads	Params (B)	Allocation rule	D4
V-width	28	224	2.442	params (FFN trim)	FAIL
V-depth	21	168	2.409	7 layers dropped	FAIL
V-depth-lite	24	192	2.414	4 layers dropped	PASS
V-depth-alt	21	168	2.409	7 layers, alt. positions	FAIL
V-KV	28	112	2.398	KV heads 8→4	FAIL
V-KV-lite	28	168	2.442	KV heads 8→6	FAIL*
V-mixed	23	92	2.420	byte-greedy mix	FAIL
V-shallow	26	208	2.398	2 top layers dropped	FAIL
V-kv2	28	56	2.420	KV heads 8→2	FAIL
V-mid	23	138	2.402	5 mid layers, KV 8→6	FAIL
V-front	24	192	2.414	4 front layers dropped	FAIL

Table 1: The eleven allocations. V-width is the parameter-count candidate and V-mixed, predeclared before measurement, is the byte-traffic candidate. Parameter counts verify the matched budget (2.398–2.442B, 1.84% spread). D4 is the amended validity verdict (Section 4.1). *V-KV-lite: the original build was export-invalid (top-1 0.000); the amended rebuild fails D4 on KL by 0.005.

The traffic model prices a decode step at occupied context c as $B(c) = B_{\text{weights}} + B_{\text{KV}}(c)$, with $B_{\text{KV}}(c) =$

$2(\sum \text{KV-heads}) \cdot 128 \cdot 2\text{B} \cdot c$. For the unpruned base this gives KV shares of total traffic of ≈24% at 4k, 39% at 8k, and 56% at 16k context — the long-context regime is where a traffic model has room to differ from parameter count, so contexts below 4096 are reported but the registered contest uses $C \geq 4096$.

3.2 Protocol

The protocol, criteria, and analysis definitions were pre-registered before any stage-1 measurement (study record SPEC v0.3, 2026-10-23; runtime amendments were logged in the same record and are summarised where they bear on the results). Timing uses llama.cpp (`llama-server`) on a 2-core virtual machine (2 vCPUs of an AMD EPYC 9D25 host). Thread caps are llama.cpp thread counts ($-t$ 2 and $-t$ 4) under a `taskset -c 0,1` pinning to two cores; "cap 4" is four threads on two cores — an oversubscribed cap, not a four-core result — labelled as such throughout. Each measurement cell (variant × cap × context ∈ {512, 2048, 4096, 6144, 8192, 16384}) runs in a fresh server process ("virgin" protocol, adopted after a slot-history contamination artifact was caught and the 30 pre-protocol reps excluded), with prefill-only primes, 5 warmup and 5 recorded reps of 128 decode steps on fixed token IDs. All decoding is single-sequence (batch size 1); the primary metric is CPU seconds per token, with per-rep host load and context switches recorded. In total 521 virgin reps are banked, of which 451 belong to the six primary variants (see Section 4.1); the contest fits use the $C \geq 4096$ subsets (135 reps at cap 2, 136 at cap 4).

3.3 Pre-registered criteria

Three frozen tests decide the claim, with one descriptive analysis alongside them. (i) *Prediction contest:* regressions of log CPU-s/token on either parameter count or modeled byte traffic (each with a layer-count term), judged on leave-one-variant-out (LOVO) relative error over $C \geq 4096$ reps; the traffic model wins only if it has lower LOVO error *and* ΔAIC (parameters − traffic) > 10. LOVO relative error is the mean, over the held-out variant's reps, of $|\text{pred} - \text{obs}|/\text{obs}$; every held-out variant is an extrapolation to an unseen architecture, so absolute values are large for both models and the contest is comparative by design. (ii) *Registered hurdle:* in genuine temporally-paired blocks (five blocks per cap × {8192, 16384}, adjacent fresh-process cells, randomised order), the upper 95% confidence bound of the paired log-time ratio V-mixed/V-width must be < 0.90 at both contexts per cap. (iii) *Validity (A3):* every variant's exported GGUF is checked teacher-forced against its own HF pruned twin (144 positions); a variant failing the frozen margins is excluded from primary fits, with a fit including the failed variants as sensitivity (Section 4.1). (iv) *Selection regret* (post-hoc and non-registered; descriptive only, computed at manuscript stage from banked data): for each cell, regret is a selector's mean time minus the fastest primary mean, relative to that mean. The amended campaign (Amendments 1–3, 2026-11-04) adds four variants, a quality evaluation, the all-variant validity check, equivalence/power analyses, a FLOP-count comparator arm, and a multicore timing arm. The amended timing (Kaggle, a public cloud-compute platform; campaign K2) did not complete: the run was cancelled at Kaggle's 12-hour session cap mid-way through the fourth variant, and the harvested partial measurements cover 46 of 120 cells over 4 of 10 variants (no verdict file was produced). The analyses consuming the amended timing (the including-fit sensitivity, TOST, power, the no-16384 secondary, and the FLOP arm, Amendment 2 item B1) therefore remain unrun and are the named remaining shortfall, outside this submission's scope; no substitute data are used for them. Amendment 4 (2026-11-05) corrects the quality scorer; see Section 4.6.

4 Results

4.1 Validity: 1 of 11 exports survives

Variant	HF NLL	dNLL	KL	top-1	D4
BASE	2.968	−0.009	0.200	0.951	ref
V-width	8.498	+0.003	0.446	0.660	FAIL
V-depth	4.376	+0.074	0.275	0.868	FAIL
V-depth-lite	3.818	+0.005	0.263	0.889	PASS
V-depth-alt	4.225	−0.011	0.468	0.826	FAIL
V-KV	7.534	−0.032	0.937	0.681	FAIL
V-KV-lite	3.614	+0.012	0.404	0.903	FAIL
V-mixed	8.488	+0.102	0.860	0.667	FAIL
V-shallow	4.576	−0.064	0.389	0.833	FAIL
V-kv2	7.811	−0.165	1.043	0.743	FAIL
V-mid	4.207	+0.024	0.590	0.833	FAIL
V-front	4.752	+0.008	0.638	0.826	FAIL

Table 2: Teacher-forced validity (A3; 144 positions; Kaggle campaign K1). dNLL, KL and top-1 compare each variant's GGUF against its own HF pruned twin, so D4 tests the export, not the pruning: a variant fails D4 when its GGUF diverges from its HF twin (quantisation divergence), whatever its pruning damage (the HF NLL column). (failing variants' KL is 1.4–5.2× BASE's 0.200.) Limits: dNLL ≤ 0.041 (one-sided: dNLL(V) ≤ dNLL(BASE) + 0.05), KL ≤ 0.399, top-1 ≥ 0.851. Two failures are marginal (V-KV-lite KL by 0.005; V-shallow top-1 by 3 positions). All 11 verdicts were recomputed from the harvested K1 files and match; the K1 raw-output pull has since landed and satisfies the quality-output provenance (Section 4.6).

The check runs on Kaggle (campaign K1; pinned base snapshot and llama.cpp 836d571; 7,956 s); the seven original variant GGUFs are byte-identical (sha256) to the test-9 clean-room rebuilds, whose packed bytes match the frozen analysis exactly. The K1 BASE GGUF differs from the

frozen BASE by 128 bytes (presumed metadata; a tensor-byte comparison is not possible from the raw pull, which contains no .gguf files), so transfer of the K1 verdicts to the frozen contest's artifacts is conditional on that comparison; the verdicts are internal to K1's own BASE reference and unaffected. The dNLL bound is one-sided by the frozen rule (a GGUF scoring better than its HF twin passes), so V-shallow (-0.064) and V-kv2 (-0.165) fail D4 on top-1 and KL, not dNLL. The 1-of-11 count is margin-sensitive: KL $\leq$ 0.45 admits V-KV-lite (2 of 11); relaxing all three limits ($|\text{dNLL}| \leq 0.07$, KL ≤ 0.45, top-1 ≥ 0.83) admits V-shallow too (3 of 11); that most variants fail is robust to these relaxations. Two readings matter. First, the D4 rule as frozen is binary and A3 applies it to primary-fit membership: the primary fit set is {V-depth-lite} alone, so the 10-variant contest cannot be fitted as a primary analysis, and a fit including all variants is the sensitivity analysis. Second, the pattern is itself the result: pruning damage (HF NLL) and quantisation divergence move together — the four variants with the worst pruning damage (V-width, V-mixed, V-kv2, V-KV; NLL $+4.6$ to $+5.5$) are also the four with KL 2.2–5.2$\times$ BASE's. The registered hurdle's two arms both fail D4; its timings are retained per the frozen rule, reported as timings of artifacts diverging from their intended pruned models beyond tolerance. V-KV-lite deserves a correction: its stage-1 invalidity (top-1 0.000, a converter bug) described the original build; the amended rebuild gives top-1 0.903 and dNLL $+0.012$ — valid in all but a marginal KL failure (0.404 against 0.399), so the amended verdict is a marginal failure, not an invalid export; the original exclusion was correct at the time.

4.2 Registered hurdle: not met (both arms invalid)

Cap	Ctx	Paired ratio	95% CI	Traffic-predicted
2	8192	0.917	[0.824, 1.020]	0.771
2	16384	0.986	[0.934, 1.041]	0.673
4	8192	0.818	[0.795, 0.842]	0.771
4	16384	0.817	[0.609, 1.096]	0.673

Table 3: Paired V-mixed/V-width time ratios (five retained pairs per cell, from the paired-block manifest). The traffic model's predicted ratio is cap-independent (it prices bytes, not threads). The hurdle (upper bound < 0.90 at both contexts) is met in neither cap: cap 2 fails: the upper bounds exceed 0.90 and the intervals include 1; cap 4 passes at 8192 but is too imprecise at 16384. Caveat: both arms fail D4 (Section 4.1); these are timings of the exported artifacts, not of validated pruned models.

At 4 threads the byte candidate wins ($\approx$0.82); at 2 threads it does not (0.92/0.99, CIs include 1). Nowhere does it meet the registered standard — and both arms fail D4, so these paired ratios compare exported artifacts that diverge from their intended pruned models beyond tolerance, not validated pruned models.

4.3 Prediction contest: traffic does not win (frozen, caveated)

Cap	Params err	Traffic err	ΔAIC (P$-$B)
2	0.743	0.689	$+2.21$ (needed > 10)
4	0.284	0.538	-19.11 (params win)

Table 4: Held-out LOVO prediction error, $C \geq 4096$ (135 reps at cap 2, 136 at cap 4), frozen six-variant primary fit. Including V-KV-lite (invalid in stage 1) as a sensitivity gives the same outcome (ΔAIC $+3.37$ at cap 2, -19.56 at cap 4). Caveat: five of these six stage-1-valid variants fail the amended (D4) validity check on rebuilt GGUFs (Section 4.1).

At 2 threads the traffic model's LOVO edge falls far short of the registered ΔAIC margin; at 4 threads the parameter model wins outright, on both criteria. The unscaled raw-byte predictor is far worse (held-out relative error 4.34 at cap 2, 4.97 at cap 4), so the contest is between the fitted models — and even fitted, the traffic model fails to justify its extra structure. With only six stage-1-valid variants, these point estimates carry wide uncertainty, quantified by a cluster bootstrap over variants (2,000 resamples): cap 2 ΔAIC median -0.91, 95% interval $[-28.8, +37.3]$, $P(\Delta\text{AIC} > 10) = 0.16$; cap 4 median -8.90, interval $[-138.4, +7.29]$, $P(> 10) = 0.02$. Read the verdict for what it is: a verdict under the registered criteria, no significance claim attached. At cap 2, nothing shows the traffic model better by anything approaching the registered margin; at cap 4, point estimates and bootstrap both favour parameters. A post-hoc sensitivity (requested in the 2026-11-04 panel review; not part of the frozen criteria) excludes the saturated 16384 context, refitting the identical contest on $C \in \{4096, 6144, 8192\}$ (98/95 reps). The cap-2 outcome is unchanged (ΔAIC $+3.70$, still below the > 10 margin), but the cap-4 outcome reverses: traffic LOVO 0.207 vs params 0.273, ΔAIC $+20.27$ — the traffic model wins there. None of this touches the frozen verdict; the sensitivity localises the failure to the longest context, where saturation and contention peak. Whether it is a true context-scaling effect or contention noise at 16384, the banked data cannot say. The amended 10-variant contest, the FLOP-count comparator arm (Amendment 2 item B1), and the equivalence/power analyses (Amendments 1–2, A4) consume the amended timing (campaign K2, cancelled incomplete; the protocol note in Section 3); under A3 their primary set is {V-depth-lite}, so they would be reported as fits including all variants, not as the pre-registered primary contest. They remain unrun, and no substitute data are used.

Model	Cap	β_0	β_1	β_2	β_3	R^2
M_P	2	-6.638	2.761	0.890	-0.347	0.412
M_B	2	12.884	-8.433	0.519	—	0.413
M_P	4	3.273	-1.287	-0.365	0.167	0.608
M_B	4	5.790	-3.628	0.383	—	0.542

Table 5: Fitted models ($y = \log$ CPU-s/token). M_P: $y \sim 1+P+S+P\cdot S+L$ ($\beta_1 = P$ in 10^9, $\beta_2 = S = (ctx+64)/10^3$, $\beta_3 = P \cdot S$); M_B: $y \sim 1 + B_W + B_{KV}(S) + L$ ($\beta_1 = B_W$ in 10^9 bytes, $\beta_2 = B_{KV}$). Layer coefficient $\approx$0 in all four fits (M_P cap 2: 0.001; M_B cap 2: 0.008; M_P cap 4: -0.003; M_B cap 4: -0.003). Coefficients are not interpreted: signs flip across caps (collinearity in a six-variant fit). Omitted from the traffic model: KV writes, activation traffic, dequantisation cost, kernel efficiency. AIC: cap 2 -328.58 (P) vs -330.78 (B); cap 4 -507.04 (P) vs -487.94 (B).

4.4 Absolute times

Absolute times run 0.22–1.38 CPU-s/token. V-depth-lite, the only D4-valid variant, is slowest at most cap-2 cells (1.376 CPU-s/token at 16384, $\pm$1.164), so validity and speed do not coincide; the fastest cells belong to V-mixed and V-depth-alt. Three cells are variance-inflated (caption); the paired ratios of Section 4.2 govern the two-candidate comparison.

Variant	Cap 2 (CPU-s/token)				Cap 4 (CPU-s/token)			
	4096	6144	8192	16384	4096	6144	8192	16384
V-width	0.257	0.299	0.558	0.818	0.250	0.310	0.373	0.766
V-depth	0.254	0.474	0.415	1.187	0.244	0.500	0.372	0.983
V-depth-lite	0.547	0.520	0.457	1.376	0.269	0.332	0.414	0.671
V-depth-alt	0.261	0.346	0.397	0.619	0.244	0.309	0.379	0.592
V-KV	0.262	0.323	0.380	0.918	0.245	0.303	0.355	0.669
V-mixed	0.230	0.280	0.415	0.655	0.219	0.255	0.310	0.625

Table 6: Cell means, CPU seconds per token ($n = 5$ reps per cell), frozen local timing. 95% CIs are in the supplementary records; three cells are variance-inflated (V-width cap-2 8192, $\pm$0.338; V-depth cap-2 6144, $\pm$0.164; V-depth-lite cap-2 16384, $\pm$1.164); the paired ratios of Section 4.2 govern the two-candidate comparison; contexts below 4096 are in the supplementary records; the amended campaign's partial cells are likewise in the supplementary records (campaign K2, incomplete).

4.5 Selection regret (descriptive)

Regret against the measured fastest primary variant, per cell (frozen cell means): V-mixed's regret is 0% at 4096–6144 (both caps) and at cap-4 8192, reaching 9.4% at cap-2 8192 (fastest V-KV) and 5.5–5.8% at 16384 (fastest V-depth-alt). V-width's regret is 6.6–47.0%. Mean regret over $C \geq 4096$ is 3.8% (V-mixed) vs 24.4% (V-width) at cap 2 and 1.4% vs 21.3% at cap 4. V-depth-alt, fastest at 16384, is one of the six primary variants, not an amendment variant. Regret rewards picks rather than models, and V-mixed also removes five layers, so its speed cannot be credited to its traffic advantage alone. The registered verdict rests on Sections 4.2–4.3; regret shows the practical stakes were asymmetric: following the traffic model would have done no harm *here* — but a model that predicts neither held-out allocations nor its own margin gives no reason to expect a repeat elsewhere.

4.6 Model quality: frozen scorer void; amended re-score reported

The A2 quality evaluation ran on Kaggle for all 12 models (40 committed tasks: extraction, arithmetic, format, summarise; greedy decoding); its frozen scores are void and are cited nowhere as evidence. The defect is in the instrument, not in any model: tasks were served by raw completion with no chat template and no stop strings, so the instruct model answers and then continues generating, while the frozen scorer assumes the output ends at the answer — "exact" compares the whole output, "number" takes the last integer, and the JSON-array match is greedy across the continuation. On BASE the first token is correct on 12/12 extraction tasks and the first number on 8/10 arithmetic tasks, yet the frozen scorer records 0/12 and 0/10 (its 14 of 40 passes all come from format and summarise); a local re-run reproduces Kaggle's frozen 35.0 exactly, so the defect is in the design, not the environment. Amendment 4 (2026-11-05) froze a first-answer re-scoring rule (first line for exact answers, first integer, first paragraph for word-count/keywords, first bullet block, non-greedy JSON match), now executed on the retrieved frozen K1 outputs (the raw pull landed; re-scoring the pulled outputs with the frozen scorer reproduces every stored score exactly). BASE scores 85.00 (34/40); the sole

Model	Amend.	Frozen	Pass	Ext/Ari/Fmt/Sum (%)
BASE	85.00	35.00	34/40	91.7/80.0/70.0/100.0
V-depth-lite	55.00	25.00	22/40	91.7/10.0/20.0/100.0
V-shallow	35.00	32.50	14/40	75.0/0.0/20.0/37.5
V-depth	30.00	7.50	12/40	75.0/0.0/0.0/37.5
V-KV-lite	27.50	12.50	11/40	33.3/10.0/30.0/37.5
V-front	7.50	5.00	3/40	0.0/0.0/0.0/37.5
V-KV	2.50	5.00	1/40	0.0/0.0/0.0/12.5
V-mid	2.50	2.50	1/40	0.0/0.0/0.0/12.5
V-depth-alt	0.00	0.00	0/40	0.0/0.0/0.0/0.0
V-kv2	0.00	0.00	0/40	0.0/0.0/0.0/0.0
V-mixed	0.00	0.00	0/40	0.0/0.0/0.0/0.0
V-width	0.00	0.00	0/40	0.0/0.0/0.0/0.0

Table 7: Amended first-answer re-score (Amendment 4) of the frozen K1 outputs, mean points of 100, beside the void frozen scores. Quality is a context axis only, never a verdict criterion.

D4-valid variant, V-depth-lite, is also the highest-scoring pruned variant (55.00); four variants score 0.00 even under first-answer scoring. The earlier "37/40" figure was a hand diagnostic on locally regenerated outputs under looser readings, superseded as a number by this re-score (the verbatim rule gives 34/40 on the K1 outputs; the reconciliation

is itemised in the supplementary record). The corrected rule still detects genuine errors — for BASE, ari02, ari06 and fmt09, plus ext05 and fmt05 under the rule as frozen — so it is not vacuous. No speed–quality frontier is claimed on one 40-task suite.

4.7 Diagnostics and saturation

Fresh diagnostic remeasurement of the earliest pool cells did not reproduce their original slowness: those cells were measured in the run's highest-contention window (host load 8–10), a load-era effect noted as a caveat. At 16384, V-depth-alt and V-mixed saturate at cap 4 (wall-clock tokens/s fall vs cap 2); their cap comparisons there are saturation-regime numbers. The head-to-head contrast is load-matched by construction (adjacent paired blocks, §4.2); no post-hoc covariate adjustment was performed (none was pre-registered).

5 Discussion

One distinction organises everything below: *selection regret is not model accuracy.* Section 4.5 shows the byte-greedy pick was, in hindsight, a low-regret pick; this section is about why that does not rehabilitate the traffic model. A model earns its keep by predicting. It predicted neither held-out allocations (Section 4.3) nor its own margins, which it overshot (Section 4.2). The parameter model should not be oversold either: absolute LOVO errors are large for *both* models (0.28–0.74 across caps), so the honest summary is a weak baseline and a challenger that adds no reliable advantage over it. A good pick justified by an inaccurate model is luck wearing a plausible story. Five candidate mechanisms follow, offered as explanations to test — rescues, they are not. **(1) Parameters already price most of the traffic.** At fixed precision, weight bytes scale as a constant multiple of the parameter count, and weights dominate traffic except at the longest contexts. The KV term — the traffic model's only independent signal — varies less across our allocations than everything else about them. Even at 16384, where KV reaches $\approx$56% of modeled base traffic, the predicted advantage (0.673) missed the observed ratio (0.986) widely. **(2) Logical bytes are not measured traffic.** Cache residency, tensor layout, and operator fusion all change actual memory movement, which a tensor-byte inventory does not see [Zhao et al., 2025]. **(3) Kernel efficiency is shape-dependent.** Cut channels or heads and matrix shapes change, attained efficiency changes with them — work volume is the least of it. That is why measured-latency methods price whole configurations [Kurtic et al., 2023] and roofline predictors need learned efficiency terms [Yao et al., 2026]. **(4) Pruning damages quantisation.** The variants whose pruned HF models are most damaged (NLL +4.6 to +5.5) are the ones whose GGUFs diverge most from their HF twins (KL 2.2–5.2$\times$ BASE's). A byte-pricing model assumes the exported model is the pruned model; for 10 of 11 allocations here, it is not. Finally, **(5) Thread saturation on a small, oversubscribed pin.** With 4 threads on 2 pinned cores, scheduling and contention move times by more than the modeled traffic differences in several cells. The evidence for saturation is narrow (two variants at 16384, §4.7); a contributing mechanism is all we claim. Practical guidance is narrow: for a one-off pruning decision, measure candidate latencies directly (as HAPE and ZipLM do); between the two analytical models, parameter count is cheaper and, here, no worse.

6 Limitations

This is a single model family (Llama-3.2-3B), a single runtime (llama.cpp, Q4_K_M, fp16 KV), a single shared 2-core virtual machine with substantial tenant load variance (per-rep load recorded; no covariate adjustment pre-registered), and two thread caps on that pin, which may mask traffic-model benefits a larger multicore setup would show. Only one of eleven variants passes the amended export-validity check, so the allocation contest cannot be run on a D4-valid primary set; the frozen contest and hurdle are reported with that caveat. Model quality is reported only as a context axis: the frozen quality instrument is void, and its Amendment-4 first-answer re-score (Section 4.6) supports no speed–quality frontier claim. The amended-timing analyses (including-fit, TOST, power, FLOP arm) remain unrun: campaign K2 was cancelled incomplete. The outcome is CPU time, not energy (no energy arm was possible in-environment: no RAPL exposure on this VM or on Kaggle), and decode is CPU-bound at fixed thread counts. The load-era caveat (§4.7) applies to the depth allocations' original cells and runs, if anything, against them, so it gives the parameter model no advantage in the primary comparison. The pre-registration history is enumerated here: SPEC v0.3 frozen 2026-10-23; virgin protocol adopted after a contamination artifact (30 pre-protocol reps excluded); base snapshot substituted once; V-mixed-shallow dropped before measurement; three restart-split paired blocks replaced wholesale; Amendments 1–3 (2026-11-04) and Amendment 4 (2026-11-05, quality-scorer correction). Beyond this regime — other CPUs, runtimes, quantisation, batch sizes — generality is untested; the claim stops where the regime does.

7 Conclusion

The test was pre-registered and the budgets matched; under it, parameter count remained unbeaten on the frozen contest — but the tightened evidence will not let that stand as a clean negative. Ten of eleven variants fail the export-validity check — a count robust to relaxed limits (3 of 11 at most,

Section 4.1) — so the allocation contest has no D4-valid primary set; the quality axis is measured only as context (BASE 85.00, the sole valid variant 55.00, under the corrected scorer). What this study does establish, twice over, is a boundary: an analytical weight-plus-KV byte-traffic model improved neither held-out prediction nor registered selection for CPU decode of pruned Llama-3.2-3B, and all but one allocation fail the export-fidelity standard under the joint prune-and-quantise pipeline (applied jointly throughout, never separated). One qualification is localised: in a post-hoc sensitivity excluding the saturated longest context (an exclusion requested in review, not pre-registered), the traffic model wins the 4-thread contest (Section 4.3). The boundary is a context boundary — and the pipeline imposes its own: check the export first.

Statement of AI Use

The author uses the AI tools Muse (Muse Spark 1.3, Meta) for design, execution, drafting, and grammatical editing assistance; Perplexity AI for literature-discovery assistance; and Google Gemini and Groq-hosted models for independent review assistance. The author reviews, edits, and attests to the accuracy of all contents.

Data availability

All timing records, the frozen analysis, the paired-block manifest, the pre-registered specification with its amendment record (Amendments 1–4), the K1 validity and quality campaign records, and the analysis scripts are archived with the study and accompany the submission as supplementary material. Comparators: parameter count and byte traffic (run); FLOP count (Amendment 2 B1, unrun: campaign K2 was cancelled incomplete); roofline bound and second runtime (omission justified, not run); named pruning methods (design-space coverage, not re-run); energy (probed, not possible in-environment).

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